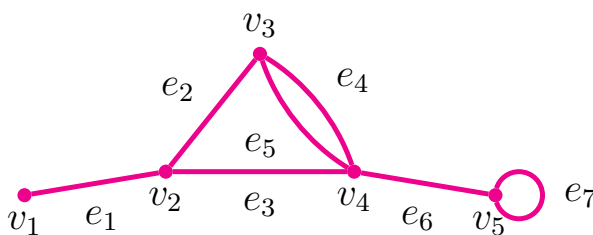


§5 Graph Theory

- Loosely speaking, a *graph* is a set of dots and dot-connecting lines.
 - The dots are called *vertices* and the lines are called *edges*.
- Formally, a graph G consists of
 - A finite set V whose elements are called the *vertices* of G ;
 - A finite set E whose elements are called the *edges* of G ;
 - A function that assigns to each edge $e \in E$ an *unordered* pair of vertices, called the *endpoints* of e .
This function is called the *edge-endpoint function*.
- Note that these graphs are **not** related to graphs of functions.
- Graphs can be used as mathematical models for networks such roads, airline routes, electrical systems, social networks, biological systems and so on.
- *Graph theory* is the study of graphs as mathematical objects.

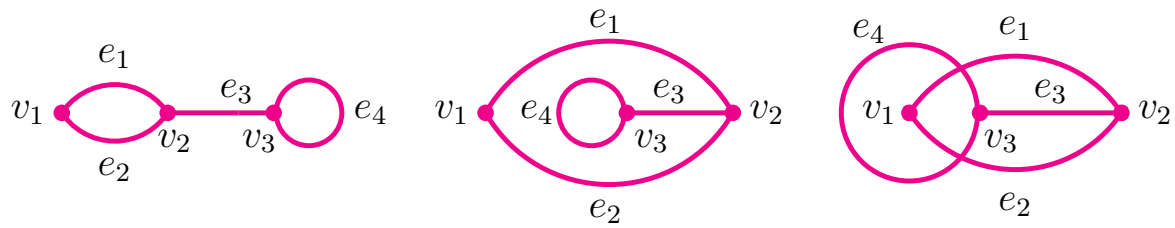
Example. Consider the following graph G with vertices and edges

$$V = \{v_1, v_2, v_3, v_4, v_5\} \quad \text{and} \quad E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\} :$$



Edge	Endpoints
e_1	$\{v_1, v_2\}$
e_2	$\{v_2, v_3\}$
e_3	$\{v_2, v_4\}$
e_4	$\{v_3, v_4\}$
e_5	$\{v_3, v_4\}$
e_6	$\{v_4, v_5\}$
e_7	$\{v_5\}$

Example. The following diagrams represent the same graph.

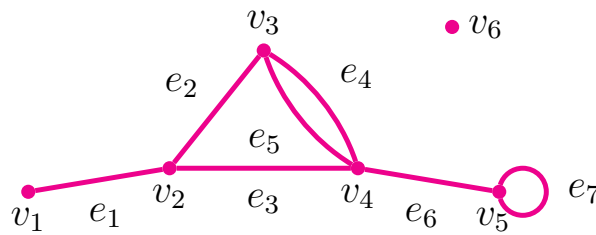


The edge-endpoint function of this graph is given below:

Edge	Endpoints
e_1	$\{v_1, v_2\}$
e_2	$\{v_1, v_2\}$
e_3	$\{v_2, v_3\}$
e_4	$\{v_3\}$

- If the edge $e \in E$ has endpoints $v, w \in V$ then we say that
 - the edge e **connects** the vertices v and w ;
 - the edge e is **incident** with the vertices v and w ;
 - the vertices v and w are the **endpoints** of the edge e ;
 - the vertices v and w are **adjacent**;
 - the vertices v and w are **neighbours**.
- Two edges **with the same endpoints** are **multiple** or **parallel**.
- A **loop** is an edge that **connects a vertex to itself**.
- The **degree** of a vertex v , denoted by $\deg(v)$, is the number of edges incident with v , counting any loops twice.
- An **isolated vertex** is one with degree 0, and a **pendant vertex** is one with degree 1.

Exercise.



Vertex	Degree
v_1	1
v_2	3
v_3	3
v_4	4
v_5	3
v_6	0

In the diagram,

- e_3 connects vertices v_2 and v_4 ;
- v_2 and v_3 are adjacent/neighbours;
- e_7 is a loop;
- e_4 and e_5 are multiple/parallel edges;
- v_1 is a pendant vertex;
- v_6 is an isolated vertex.

• The Handshaking Theorem.

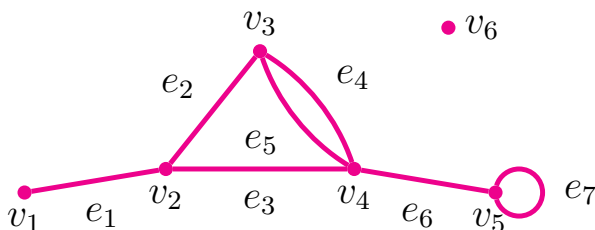
The total degree of a graph is twice the number of edges:

$$2|E| = \sum_{v \in V} \deg(v).$$

Proof.

Each edge has two endpoints and must contribute 2 to the sum of degrees, which is why we count a loop twice.

Example.



$$2|E| = 2 \cdot 7 = 14$$

$$\sum_{v \in V} \deg(v) = 1 + 3 + 3 + 4 + 3 + 0 = 14$$

- By the Handshaking Theorem, the **total degree of a graph** must be even and the **number of odd vertices** must be even.

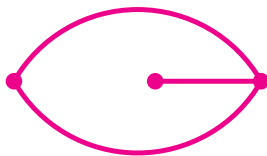
Example. No graph can have vertex degrees 3,3,3,2,2.
Why?

🔴 A *simple graph* is a graph with **no** loops or parallel edges.

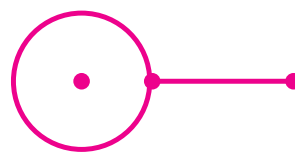
Example.



A simple graph



Not a simple graph



Not a simple graph

🔴 Note: each vertex in a simple graph on n vertices has degree at most $n - 1$.

Why?

Exercise. Prove that no simple graph can have the following vertex degrees:

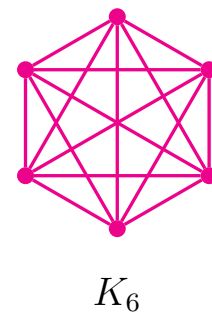
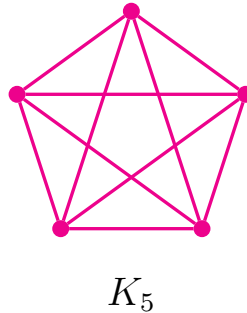
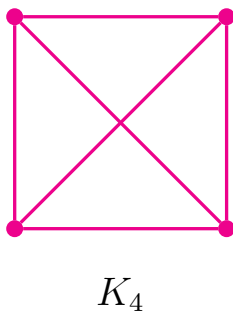
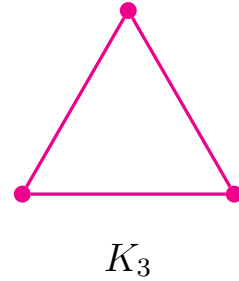
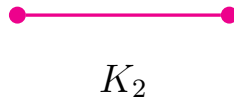
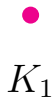
🟡 5,4,3,2,2;

🟡 4,3,3,1,1.

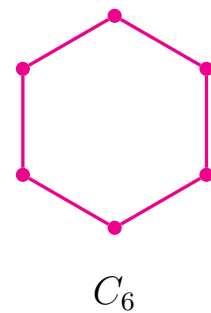
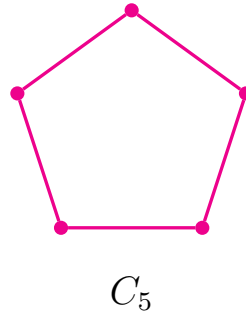
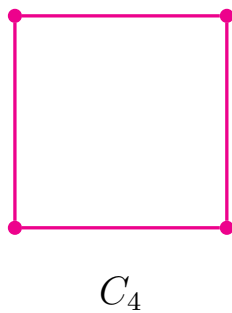
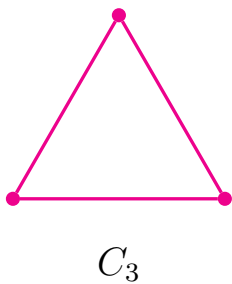
SOME NAMED GRAPHS

1. The **complete graph** K_n ($n \geq 1$) is a simple graph with
- n vertices;
 - exactly one edge between each pair of distinct vertices.

Hence K_n has $C(n, 2)$ edges.



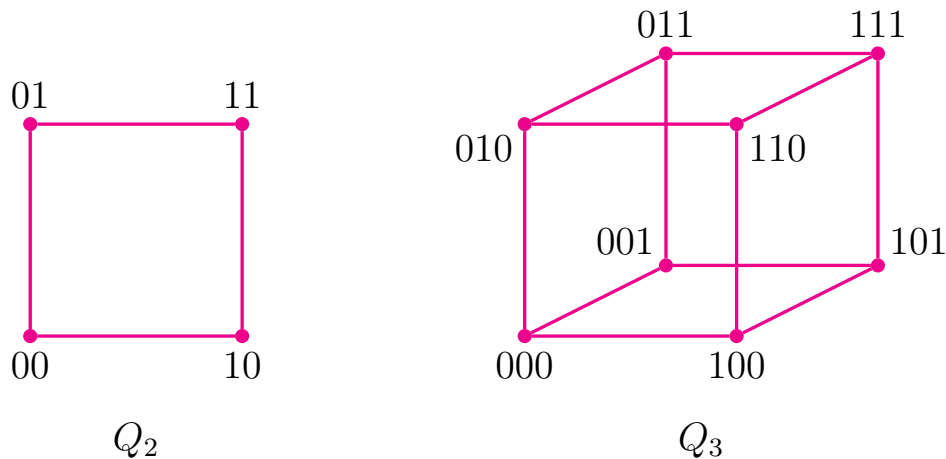
2. The **cyclic graph** C_n ($n \geq 3$) consists of
- n vertices $v_1, v_2, \dots, v_n$;
 - n edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.



3. The n -cube Q_n is the simple graph with

- vertices for each bit string $a_1a_2 \cdots a_n$ of length n , where $a_i \in \{0, 1\}$;
- an edge between vertices $a_1a_2 \cdots a_n$ and $b_1b_2 \cdots b_n$ if and only if $a_j \neq b_j$ for exactly one $j \in \{1, \dots, n\}$.

★ Two vertices are adjacent if and only if they differ by one bit.

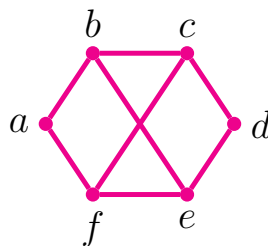


BIPARTITE GRAPHS

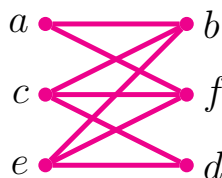
• A graph is *bipartite* iff its vertex set V can be partitioned into subsets V_1, V_2 so that every edge has an endpoint in V_1 and an endpoint in V_2 .

• In other words, no vertex is adjacent to any vertex in the same subset.

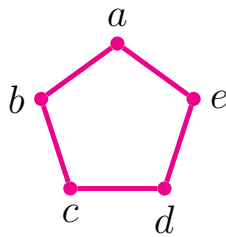
Example. Is the following graph bipartite?



Yes: let $V_1 = \{a, c, e\}$ and $V_2 = \{b, d, f\}$;
then each edge has a vertex in V_1 and a vertex in V_2 ,
as we can see by redrawing the graph, for instance as follows:



Example. Is the cycle C_5 bipartite?



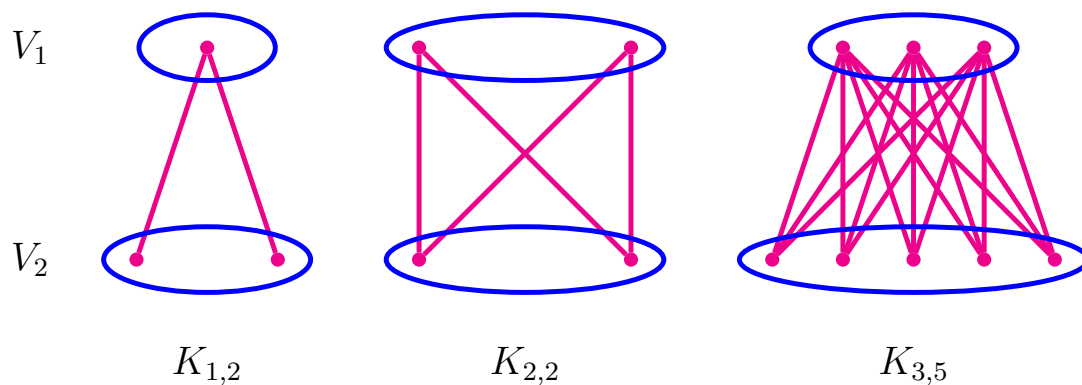
No.

● The *complete bipartite graph* $K_{m,n}$ is the simple bipartite graph with vertex set $V_1 \cup V_2$, with

- V_1 containing m vertices and V_2 containing n vertices;
- edges between **every** vertex in V_1 and **every** vertex in V_2 .

$K_{m,n}$ has $m + n$ vertices and mn edges.

Example.

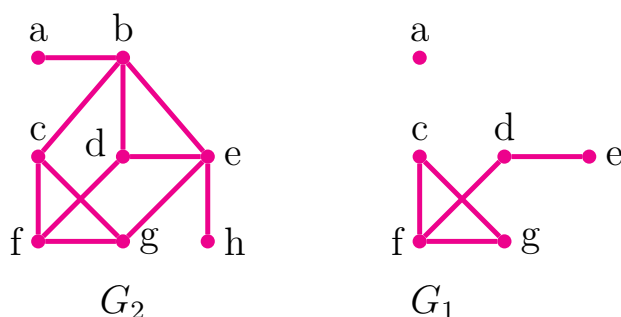


Let G_1 and G_2 be two graphs with vertex sets V_1 and V_2 and edge sets E_1 and E_2 , respectively. Then G_1 is a **subgraph** of G_2 , and we write $G_1 \subseteq G_2$, iff

- $V_1 \subseteq V_2$;
- $E_1 \subseteq E_2$;
- each edge in G_1 has the same endpoints as in G_2 .

Pictorially, a graph obtained by deleting some edges and/or vertices is a subgraph. If a vertex is deleted then all edges incident with it are also deleted.

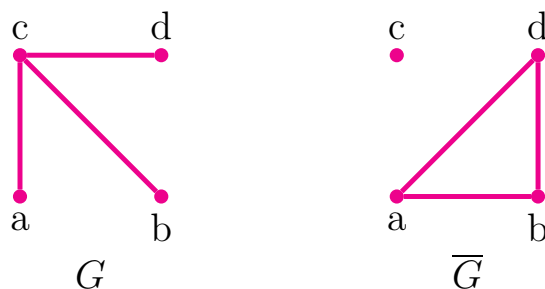
Example. $G_1 \subseteq G_2$.



Let G be a simple graph. The **complementary graph** $\overline{G}$ of G is a simple graph with

- the same vertex set as G ;
- an edge joining two vertices if and only if they are **not** adjacent in G .

Example.



Problem Set 5, Problem 10.

If a simple graph G has n vertices and m edges, then how many edges does $\overline{G}$ have?

..... Now try Problem Set 5, Questions 1–12.

ADJACENCY MATRIX

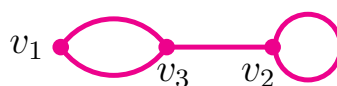
- Let G be a graph with an ordered listing of vertices $v_1, v_2, \dots, v_n$.
The *adjacency matrix* of G is the $n \times n$ matrix $A = [a_{ij}]$ with

$$a_{ij} = \# \text{ edges connecting } v_i \text{ and } v_j.$$
- The entries a_{ij} depend on the order in which the vertices have been numbered.
 - Changing the vertex order corresponds to permuting rows and columns.
- The adjacency matrix A is symmetric, i.e., $A = A^T$.

Example.



$$A = \begin{bmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

Exercise. Is the graph G with adjacency matrix A below simple?

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Answer. 0's on the diagonal of $A \implies$

All off-diagonal entries $\leq 1 \implies$

..... Now try Problem Set 5, Questions 13 and 14.

PATHS & CIRCUITS

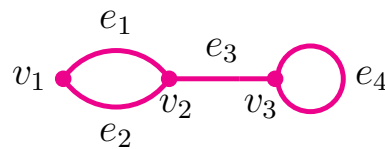
- A **walk** in a graph G is an alternating sequence of vertices v_i and edges e_i in G

$$v_0 e_1 v_1 e_2 v_2 \dots v_{n-1} e_n v_n$$

where v_{i-1} and v_i are the endpoints of edge e_i for all i .

- The **length** of the walk is the number of edges involved (n above).
- A **closed** walk is one that **starts and ends in the same vertex**.
- In a simple graph, a walk can be specified by stating the vertices alone.

Example.

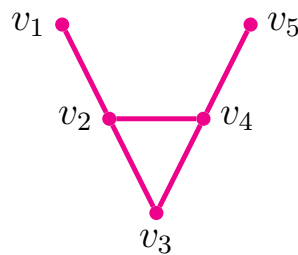


$v_1 e_1 v_2 e_3 v_3 e_4 v_3$ is a walk of length 3 from v_1 to v_3 .

$v_1 e_1 v_2 e_2 v_1$ is a closed walk of length 2 from v_1 to v_1 .

- A **path** is a **walk with no repeated edges**.
- A **circuit** is a **closed walk with no repeated edges**.

Example. Consider a simple graph



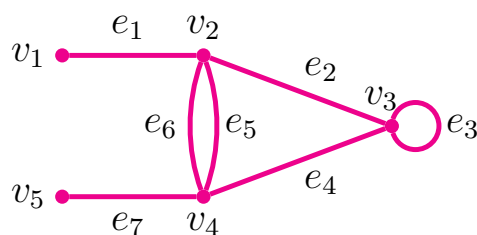
$v_1 v_2 v_4 v_3 v_2$ is a walk of length 4 from v_1 to v_2 . It is also a path.

$v_1 v_2 v_4 v_3 v_2 v_4 v_5$ is a walk but not a path because edge $v_2 v_4$ is repeated.

$v_2 v_3 v_4 v_2$ is a circuit.

- A path $v_0e_1v_1e_2v_2 \dots v_{n-1}e_nv_n$ is *simple* iff all the v_i are distinct i.e. there are no repeated vertices.
- A circuit $v_0e_1v_1e_2v_2 \dots v_{n-1}e_nv_n$ is *simple* iff $v_1, \dots, v_n$ are distinct (but $v_0 = v_n$ of course).

Example.



$v_5e_7v_4e_6v_2e_2v_3e_4v_4e_5v_2$ is a path of length 5 from v_5 to v_2 . It is not simple.

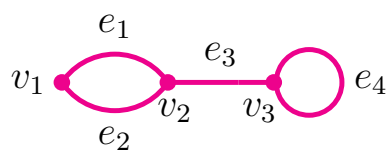
$v_1e_1v_2e_2v_3e_4v_4e_7v_5$ is a simple path of length 4 from v_1 to v_5 .

$v_2e_6v_4e_4v_3e_2v_2$ is a simple circuit of length 3.

- **Theorem.** Let a and b be vertices in a graph.
If there is a walk from a to b , then there is a simple path from a to b .

- **Theorem.** If A is the adjacency matrix for G with ordered vertices $v_1, \dots, v_n$, then the number of walks of length k from v_i to v_j in G is given by the entry in the i th row and j th column of A^k .

Example.



$$A = \begin{bmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad A^2 = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

From A^2 we can see that there are

🟡 2 walks of length 2 from v_1 to v_3 , namely

$$v_1 e_1 v_2 e_3 v_3 \quad \text{and} \quad v_1 e_2 v_2 e_3 v_3;$$

🟡 4 walks of length 2 from v_1 to v_1 , namely

Proof. (of the Theorem)

..... Now try Problem Set 5, Questions 19 and 20.

CONNECTIVITY

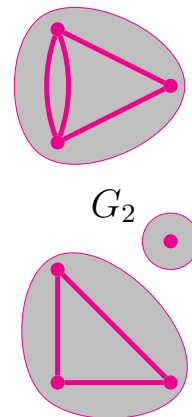
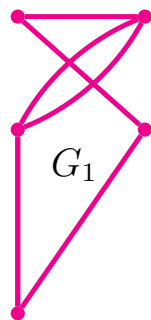
- Vertices a, b of a graph G are **connected** in G iff there is a walk from a to b .
- A graph G is **connected** iff every pair of distinct vertices is connected in G .
- Let G be a graph with vertex set V .
The relation $\sim$ on V defined by

$$v_i \sim v_j \text{ if and only if } v_i \text{ is connected to } v_j \text{ in } G$$

is an equivalence relation.

- The equivalence classes of this relation are the **connected components of G** .
Two vertices are in the same connected component if and only if they are connected in G .

Example. The graph G_1 below is connected, but G_2 is disconnected. The 3 connected components of G_2 have been shaded.



- Let G be a graph with vertices $v_1, \dots, v_n$ and adjacency matrix A . Let

$$C = I + A + A^2 + \dots + A^{n-1}.$$

Then G is connected if and only if C has **no zero entries**.

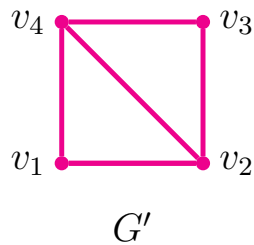
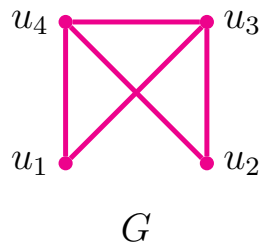
Proof.

.....Now try Problem Set 5, Question 18.

ISOMORPHIC GRAPHS

- Let G and G' be graphs with vertices V , resp. V' , and edges E , resp. E' . Then G is **isomorphic** to G' , and we write $G \simeq G'$, iff there are **two bijections** $f : V \rightarrow V'$ and $g : E \rightarrow E'$, such that **e is incident with v in G if and only if $g(e)$ is incident with $f(v)$ in G'** .
 - Roughly speaking, two graphs are **isomorphic** iff they are the **same except for edge and vertex labelings**.
 - In this case, $\deg(v) = \deg(f(v))$.
- Two **simple** graphs G and G' are isomorphic iff there is a bijection $f : V \rightarrow V'$ such that for all $v_1, v_2 \in V$, **v_1 and v_2 are adjacent in G if and only if $f(v_1)$ and $f(v_2)$ are adjacent in G'** .

Exercise. Are the following simple graphs isomorphic?



v	$f(v)$
u_1	v_1
u_2	v_3
u_3	v_4
u_4	v_2

Yes; an isomorphism is given by the table.

Note that u_i and u_j are adjacent if and only if $f(u_i)$ and $f(u_j)$ are adjacent.

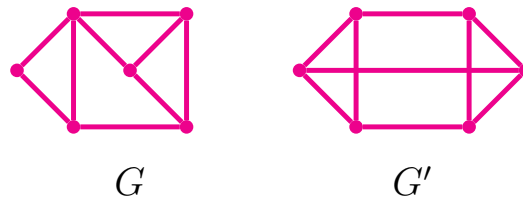
• A property of a graph G is an *invariant* iff G' also has this property whenever $G' \simeq G$.

Example. Some graph invariants are

- the number of vertices;
- the number of edges;
- the total degree;
- the number of vertices of a given degree;
- bipartiteness, number of connected components, connectedness;
- having a vertex of some degree n adjacent to a vertex of some degree m ;
- the number of circuits of a given length.

- The easiest way to show that graphs G and G' are **not** isomorphic ($G \not\simeq G'$) is to **find an invariant property that holds for G but not for G'** .
- To prove that simple graphs G and G' **are** isomorphic ($G \simeq G'$), we need to **find an isomorphism between them**; that is, a bijection $f : V \rightarrow V'$ satisfying the condition for isomorphism.
- If G and G' have n vertices then there are $n!$ bijections from V to V' . If n is large then it is very hard to find an isomorphism among all $n!$ bijections.

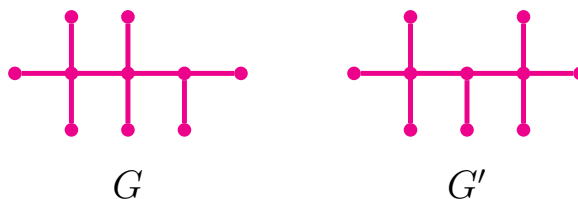
Exercise. Are these two simple graphs isomorphic?



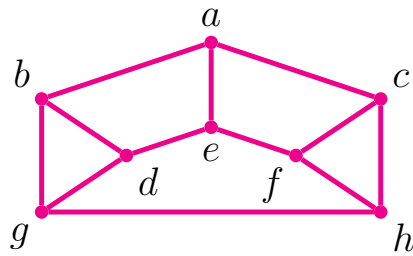
No:

G and G' are not isomorphic since G has a vertex of degree 4 but G' does not.

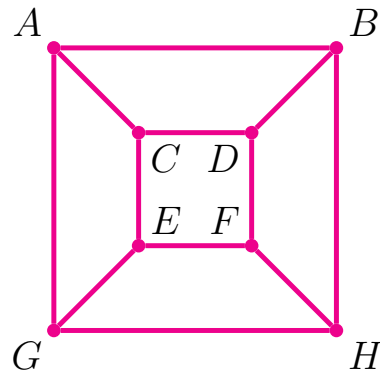
Exercise. Are these two simple graphs isomorphic?



Exercise. Are these two simple graphs isomorphic?



G_1



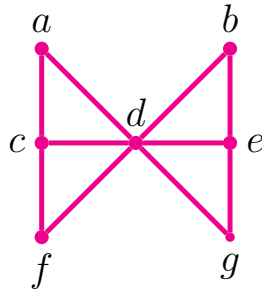
G_2

.....Now try Problem Set 5, Questions 15–17.

EULER & HAMILTON PATHS & CIRCUITS

- Let G be a graph.
- An *Euler path* in G is a path that includes every edge of G exactly once.
- A *Hamiltonian path* in G is a simple path that includes every vertex of G exactly once.
- An *Eulerian circuit* in G is a circuit that includes every edge of G exactly once.
- A *Hamiltonian circuit* in G is a simple circuit which includes every vertex of G exactly once (counting the start-vertex once).

Example.



$cfdacdgebde$ is an Euler path;

$acfdbeg$ is a Hamilton path.

Example. A network of roads is to be snow-ploughed.
An Euler circuit will ensure that all roads get ploughed
without going over a road already cleared.

Example. Similarly for a postman delivering mail.

Example. A salesperson wants to visit some towns using a network of roads.
If the salesperson follows a Hamilton circuit then they will visit each town
exactly once, and will finish up where they started.

● **Theorem.** Let G be a **connected** graph.

An Euler circuit exists if and only if G has **even vertex degrees** i.e. there are
no vertices with odd degree.

($\Leftarrow$) Conversely, suppose that G has no vertex of odd degree. We need

● **Lemma** Let G' be a graph with even vertex degrees.
Then any path $v_0 e_1 v_1 \dots e_n v_n$ with $v_0 \neq v_n$ can be extended to a circuit
 $v_0 e_1 v_1 \dots e_n v_n e_{n+1} \dots v_m$ with $v_m = v_0$.

We now return to the proof of ($\Leftarrow$) and use the following algorithm to produce an Euler circuit C :

Euler Circuit

Choose a vertex v_0 in G . Start with length 0 circuit $C := v_0$.

while there exists an edge not in C do

 choose a vertex $v \in C$ and an edge $e \notin C$ that is incident with v ;

 choose a circuit C' starting and finishing at v that contains e ,

 but does not contain any edge in C ;

 Replace one of the appearances of v in C by C' ;

end do

The vertex v and edge e exists because G is connected.

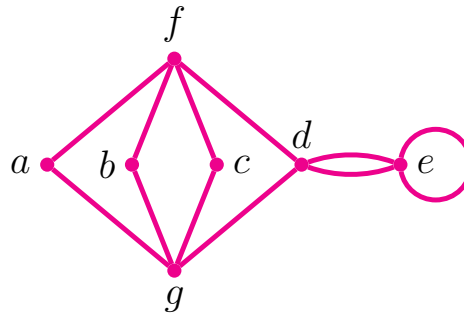
To see that the circuit C' exists, apply the Lemma to the graph obtained from G by deleting all edges of C .

The algorithm terminates because G has only finitely many edges, and every step adds at least one edge to C .

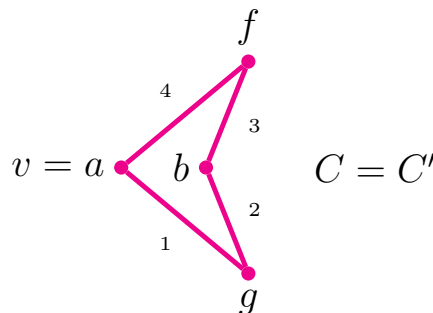
By construction, C is an Euler circuit.

This proves the theorem.

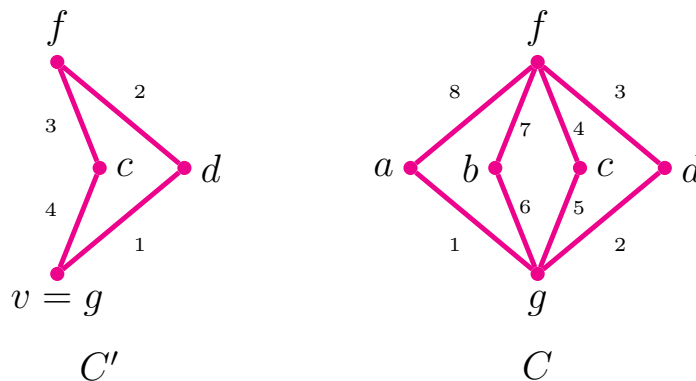
Example. Apply **Euler Circuit** to construct an Euler circuit below:



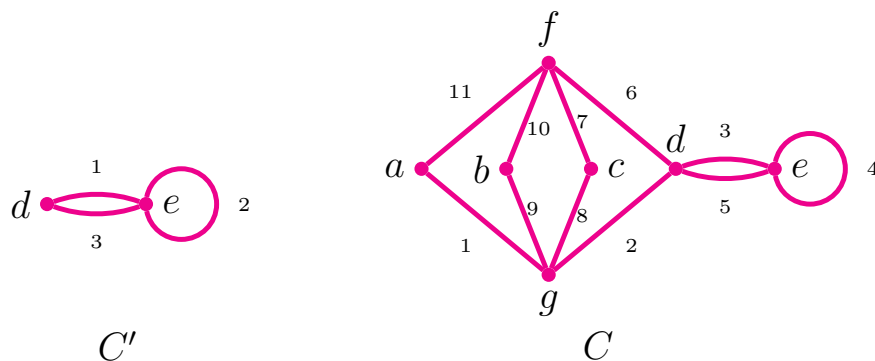
First note that the degree of each vertex is even.
 If we initially take $v_0 := a$ then at the first step of the algorithm, one possible choice for C' is as follows:



At the second step, we may take $v := g$ and choose C' as follows:



At the third step, we must take $v := d$, and can choose C' as follows:



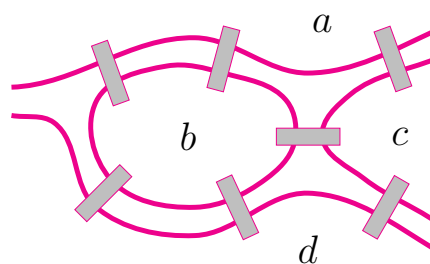
We now have an Euler circuit C .

● **Theorem.** Let G be a **connected** graph.

An Euler path which is not a circuit exists if and only if G has **exactly two vertices of odd degree**.

The Königsberg Bridge Problem.

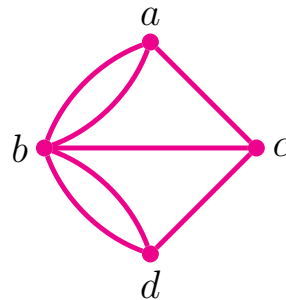
In the 18 century, Kaliningrad in the Russian Republic was called Königsberg, and was part of Prussia. At that time, seven bridges connected the four different parts of the town, as depicted in the diagram below.



The question arose as to whether it was possible to start from one part of the town, cross every bridge exactly once, and return to the starting point.

In the first ever paper on graph theory, in 1736, Leonhard Euler explained why this could not be done and proved the theorems about when an Euler circuit or Euler path exist.

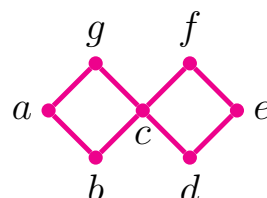
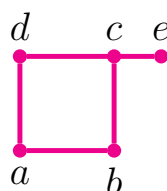
We now know that the problem is equivalent to finding an Euler circuit in the following graph.



Here, $\deg(a) = 3$, $\deg(b) = 5$, $\deg(c) = 3$, and $\deg(d) = 3$, so there are 4 vertices with odd degree, and hence no Euler circuit (or path) exists.

- No simple necessary and sufficient criteria are known that determine whether a graph has a Hamilton circuit or path.
- Note that a graph with a **vertex of degree 1** cannot have a Hamilton circuit.
- If a graph G has a Hamilton circuit, then the circuit must include all edges incident with vertices of degree 2.
- A Hamilton path or circuit uses at most 2 edges incident with any one vertex.

Example. The following graphs have no Hamilton circuits:



To see that the second graph has no Hamilton circuit, note that b , d , f , g each have degree 2, so any Hamilton circuit would have to use all edges incident with these vertices, including the four edges incident with c .

● **Theorem.** (Dirac 1952)

Let G be a **connected** and **simple** graph with $n \geq 3$ vertices, such that each vertex has degree at least $n/2$. Then G has a Hamilton circuit.

● **Example.** The complete graph K_n is connected and simple, and has n vertices.

Each vertex is adjacent to every other vertex, so each vertex has degree $n - 1$. By the theorem, K_n has a Hamilton circuit when $n \geq 3$.

● The above theorem does **not** give a **necessary** condition. Some graphs have Hamilton circuits but do not satisfy the theorem's condition.

● **Example.** The cyclic graph C_n has a Hamilton circuit for all $n \geq 3$. Each vertex has degree 2 which is smaller than $n/2$ when $n \geq 5$.

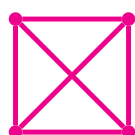
.....Now try Problem Set 5, Questions 21–26.
PLANAR GRAPHS

● A graph G is **planar**

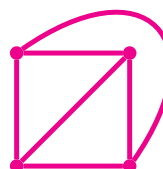
iff it **can be drawn in the plane** so that no edge crosses another.

● Such a drawing is called a **planar map** or **planar representation** of G .

Example. The complete graph K_4 is planar:



not a planar map of K_4



a planar map of K_4

Exercise. Prove that the complete bipartite graph $K_{2,3}$ is planar.

Note. We will see later that K_5 and $K_{3,3}$ are not planar.

- A planar map divides the plane into a finite number of *regions*. Exactly one of these regions is unbounded.
- A planar graph can have different planar representations (or maps), but the *number of regions* is the **same** for all planar representations. This number depends only on the number of edges and vertices of the graph:
- **Euler's Formula.**
If G is a connected planar graph with e edges and v vertices, and if r is the number of regions in a planar representation of G , then

$$v - e + r = 2.$$

Example. Consider the following planar map:



The map has $v = 3$ vertices, $e = 4$ edges, and $r = 3$ regions. Therefore, $v - e + r = 2$.

Exercise. If G is a connected planar graph with 8 vertices each with degree 3 then how many regions are there in a planar representation for G ?

• **Lemma.** Let G be a connected graph with more than one vertex.
If G has no circuit, then it has a pendant vertex.

Proof.

Proof of Euler's Formula. We will use (informal) induction on e :

- If $e = 0$, then $v = 1$ and $r = 1$ because the map is connected.
Thus, $v - e + r = 1 - 0 + 1 = 2$ as required.

- Assume for some $e \geq 0$ that the formula holds for all connected planar map with e edges. Let G be a connected planar map with $e + 1$ edges, v vertices, and r regions.

We must prove that $v - (e + 1) + r = 2$.

By induction, Euler's Formula holds for all connected planar maps.

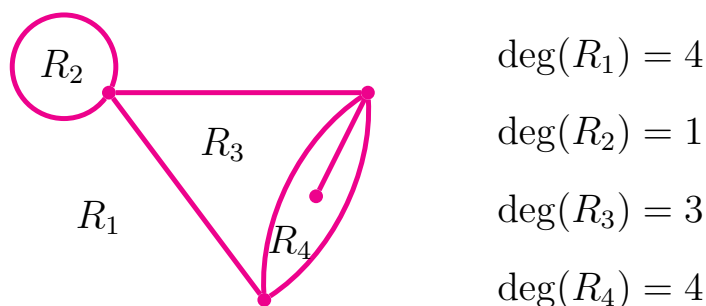
□

- The **degree** of a region R in a planar representation is the number of edges (counting repetitions) traversed in going round the boundary of R .
- In summing all region degrees, each edge contributes twice, so

$$2|E| = \text{the sum of region degrees.}$$

- By the Handshaking Lemma, it follows that
 - the sum of region degrees equals the sum of vertex degrees.

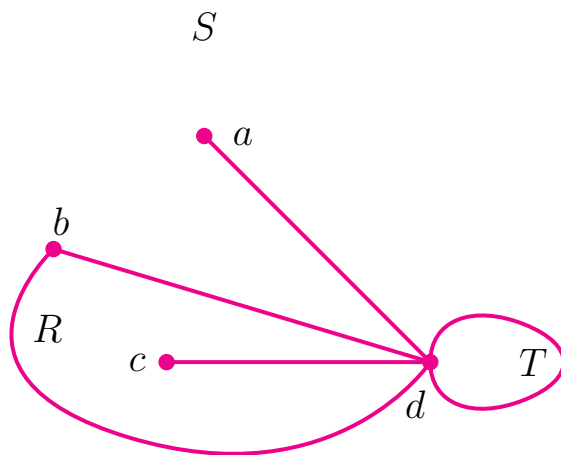
Example.



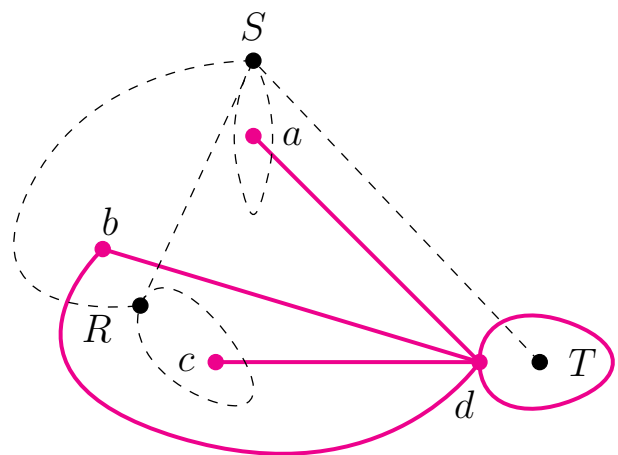
- The **dual** of a planar map G is a **graph** G^* given as follows:
 - for each region R_i of G , there is an associated vertex v_i^* in G^* ;
 - for each edge e in G that is surrounded by one region R_i , there is an associated loop in G^* at vertex v_i^* .
 - for each edge e of G that separates two regions R_1 and R_2 , there is an edge e^* in G^* that connects vertices v_1^* and v_2^* corresponding to R_1 and R_2 , respectively;
- The dual G^* is a **planar graph**.

Example.

The planar map G shown on the left has three regions, labelled R, S, T .



The planar map G



Constructing the dual graph G^*

The dual graph is



- If G is a **simple connected planar graph** with at least 3 vertices, then every region degree is at least 3.
 - To have a region of degree 1, G must have a **loop**.
 - To have a region of degree 2, G must have **parallel edges**.
- **Theorem.**
If G is a connected planar simple graph with e edges and $v \geq 3$ vertices, then
 - $e \leq 3v - 6$;
 - $e \leq 2v - 4$ if G has no circuits of length 3.
- This theorem is useful for proving that some graphs are **not** planar.
- This theorem is an example of the principle that, the more edges a graph has, the harder for it to be planar.

Proof.

Example. Prove that the complete graph K_5 is not planar.

K_5 is connected and simple,

but $e = C(5, 2) = 10$ and $3v - 6 = 3 \times 5 - 6 = 9$.

By the theorem, K_5 is not planar.

Example. Prove that the complete bipartite graph $K_{3,3}$ is not planar.

Note that a path of length 3 starting in one vertex set must end in the other.

Therefore, $K_{3,3}$ has no circuits of length 3.

Since $2v - 4 = 2 \times 6 - 4 = 8 < 9 = e$, $K_{3,3}$ cannot be planar.

Note. $3v - 6 = 3 \times 6 - 6 = 12$,

so we could not use the inequality $e \leq 3v - 6$.

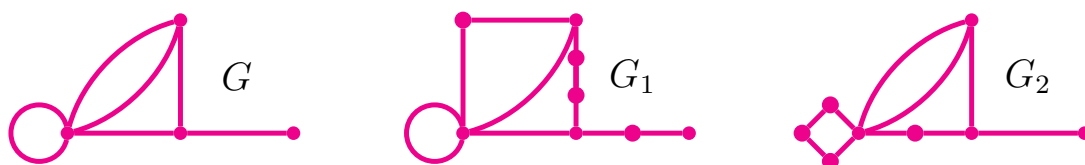
- Suppose that G has an edge e with endpoints v and w .
Let G' be the graph obtained from G by replacing e by a path $ve'v'e''w$.
- Such an operation is called an *elementary subdivision*.



- If G is planar then so is G' .
- Two graphs are *homeomorphic* iff each can be obtained from a common graph by elementary subdivisions.
 - If G is planar then so is any graph homeomorphic to G .

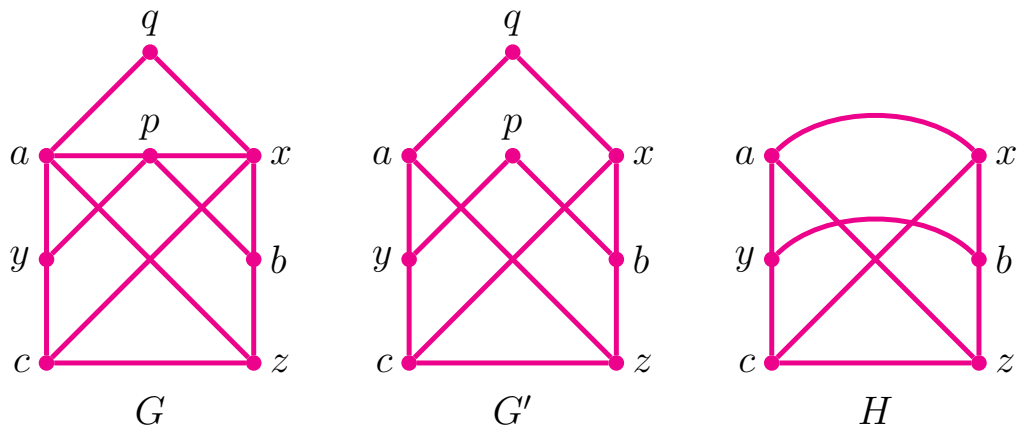
Example. Consider the graphs G , G_1 , and G_2 below.

Since G_1 and G_2 each are obtained from G by elementary subdivisions, it follows that G_1 and G_2 are homeomorphic.



- Kuratowski's Theorem.** (1930) A graph is *planar* if and only if it does not contain a subgraph homeomorphic to $K_{3,3}$ or K_5 .

Example. Show that the graph G below is not planar.



The first theorem on planarity doesn't help, since $e = 13$ and $3v - 6 = 18$, and G does have circuits of length 3. But we can use **Kuratowski's Theorem**:

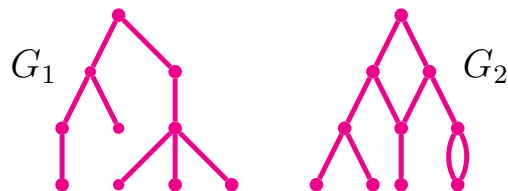
- * the subgraph G' obtained by deleting edges ap and px is homeomorphic to H (by subdivision of edges ax and yb in H),
- * H is isomorphic to $K_{3,3}$ (with vertex partition $\{a, b, c\}$ and $\{x, y, z\}$).

.....Now try Problem Set 5, Questions 27–34.

TREES

- A **tree** is a **connected graph with no circuits**.
- A tree has no loops or multiple edges so it is **simple**.

Example. The graph G_1 is a tree but G_2 is not (it has two simple circuits).

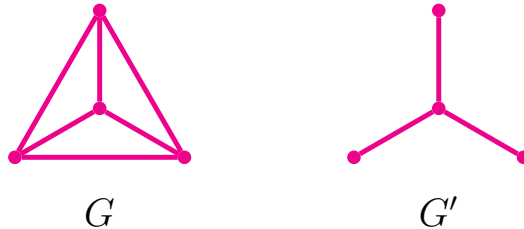


- **Theorem.** Any tree T is planar.

Why?

● A *spanning tree* in a graph G is
a subgraph that is a tree and contains every vertex of G .

Example. The subgraph G' below is a spanning tree for the graph G :



Example. Given a network of dirt roads connecting various towns, we may want to pave a minimal subset of the roads so as to ensure that every pair of towns is connected by a paved route.

To do this, we should pick a spanning tree for the network.

● **Theorem.** Every connected graph contains a spanning tree.

Proof.

● **Theorem.** A connected graph with n vertices is a tree
if and only if it has exactly $n - 1$ edges.

Proof.

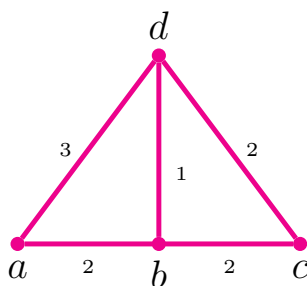
Exercise. Find an easy induction proof of this result.

Exercise.

Prove that the spanning trees of a graph all have the same number of edges.

- A *weighted graph* is a graph whose edges have been given numbers called *weights*. The *weight of an edge* e is denoted by $w(e)$.
- The *weight of a subgraph* in a weighted graph G is the sum of the weights of the edges in the subgraph.
- These numbers often represent lengths, travel time, costs, flow capacity, etc.
- A *minimal spanning tree* in a weighted graph G is a spanning tree whose weight is less than or equal to the weight of any other spanning tree.
- There can be more than one minimal spanning tree in a graph.

Example. Consider the following weighted graph.



The edges da , db , and dc form a spanning tree T of total weight 6.

This tree T is not minimal

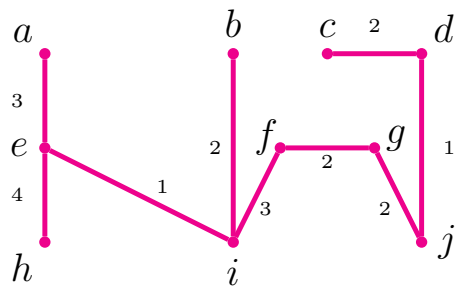
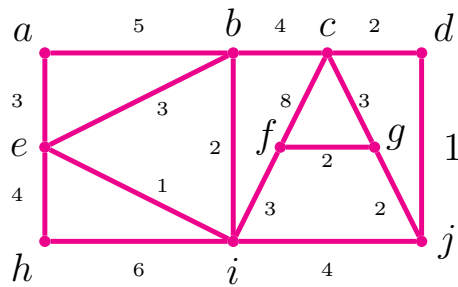
since edges ab , bd , and bc form a spanning tree T' of total weight 5.

It is not hard to see that T' is minimal.

The edges ab , bd , and dc form another minimal spanning tree T'' .

- Minimal spanning trees of weighted graphs G on n vertices are found using **Kruskal's Algorithm** (1928):
 - Start with the tree $T := \emptyset$.
 - Sort the edges of G into increasing order of weight.
 - Going down the list, add an edge to T if and only if it does not form a circuit with edges already in T .
 - Continue this process until T has $n - 1$ edges.
 - Then T is a minimal spanning tree for G .

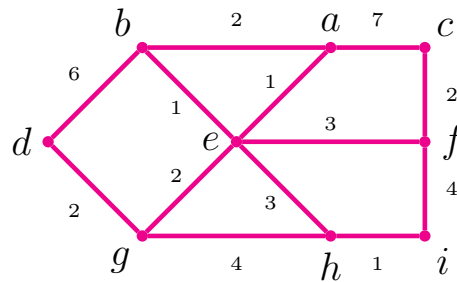
Example. Find a minimal spanning tree for the following weighted graph.



Edge	Weight	Chosen?
dj	1	Y
ei	1	Y
bi	2	Y
cd	2	Y
fg	2	Y
gj	2	Y
ae	3	Y
be	3	N
cg	3	N
fi	3	Y
bc	4	N
eh	4	Y, stop
ij	4	
ab	5	
hi	6	
cf	8	

- Given a connected weighted graph G and a particular vertex v_0 , we want to find a *shortest path from v_0 to v* for each vertex v in G (here, a *shortest path* is one with *minimal total weight*).
- The union of these paths forms a *minimal v_0 -path spanning tree* for G .
- Dijkstra's Algorithm** (1959)
 - Start with the subgraph T consisting of v_0 only.
 - Consider all edges e with one vertex in T and the other vertex v not in T .
 - Of these edges, choose an edge e giving a shortest path from v_0 to v .
 - Add this edge e and vertex v to T .
 - Continue this process until T contains all the vertices of G .
 - Then T is a minimal v_0 -path spanning tree for G .

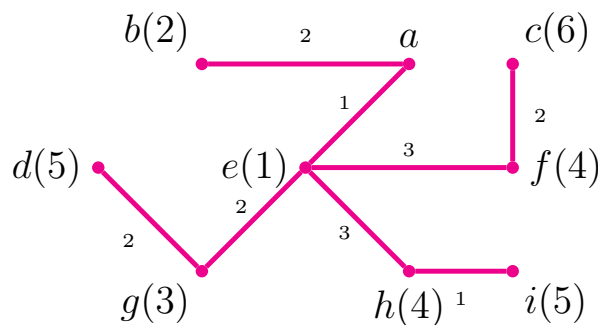
Example. Find the shortest paths from the vertex a to each of the other vertices in the weighted graph shown below (i.e., find a minimal a -path spanning tree).



There are nine vertices, so DIJKSTRA'S ALGORITHM has eight steps, given by the following table, where the total distances from a are in brackets.

Edge candidates	Next edge	Next vertex
$ab(2), ae(1), ac(7)$	$ae(1)$	$e(1)$
$ab(2), ac(7), eb(2), eg(3), eh(4), ef(4)$	$ab(2)$	$b(2)$
$ac(7), bd(8), eg(3), eh(4), ef(4)$	$eg(3)$	$g(3)$
$ac(7), bd(8), eh(4), ef(4), gd(5), gh(7)$	$eh(4)$	$h(4)$
$ac(7), bd(8), ef(4), gd(5), hi(5)$	$ef(4)$	$f(4)$
$ac(7), bd(8), gd(5), hi(5), fc(6), fi(8)$	$gd(5)$	$d(5)$
$ac(7), hi(5), fc(6), fi(8)$	$hi(5)$	$i(5)$
$ac(7), fc(6)$	$fc(6)$	$c(6)$

The tree produced by the algorithm is shown below.

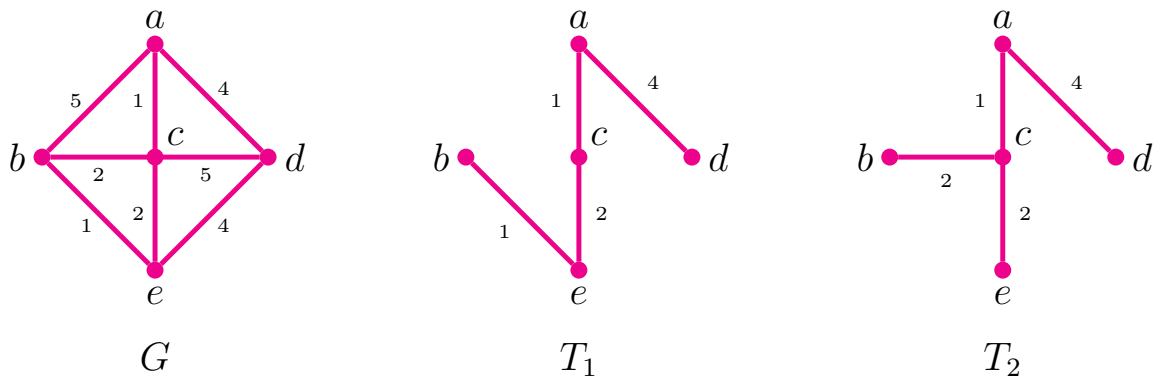


Note. At the second step, we could have chosen eb instead of ab .

• The minimal a -path spanning tree is **not** generally a minimal spanning tree.

Example.

The tree T_1 below is a minimal spanning tree in G (total weight = 8) whereas T_2 is a minimal a -path spanning tree in G (total weight = 9).



.....Now try Problem Set 5, Questions 35–40.

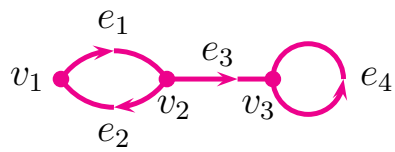
EXTRA MATERIAL NOT EXAMINED BUT IN SYLLABUS

DIRECTED GRAPHS

- Loosely speaking, a *directed graph* is a set of dots and dot-connecting arrows; in other words, a graph whose edges have been pointed one way or the other.
- Formally, a *directed graph* D consists of
 - A set V whose elements are called the *vertices* of D ;
 - A set E whose elements are called the *(directed) edges* of D ;
 - An *edge-endpoint function* that assigns to each (directed) edge $e \in E$ an *ordered* pair of vertices called the *endpoints* of e .
 - The *first vertex* of the pair is called the *initial* or *start vertex*;
 - the *second vertex* is called the *final* or *finish vertex*.

Example. Internet pages and links form the vertices and edges of a directed graph.

Exercise.

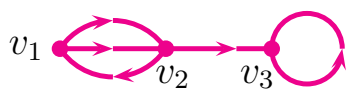


Edge	Endpoints
e_1	(v_1, v_2)
e_2	(v_2, v_1)
e_3	(v_2, v_3)
e_4	(v_3, v_3)

ADJACENCY MATRIX OF A DIRECTED GRAPH

- Let G be a directed graph with an ordered listing of vertices $v_1, v_2, \dots, v_n$. The **adjacency matrix** of G is the $n \times n$ matrix $A = [a_{ij}]$ with
 - $a_{ij} = \#$ edges with start point v_i and final point v_j ,
- The entries a_{ij} depend on the order in which the vertices have been numbered.
 - Changing the vertex order corresponds to permuting rows and columns.
- If G is directed then the adjacency matrix A need not be symmetric.

Example.



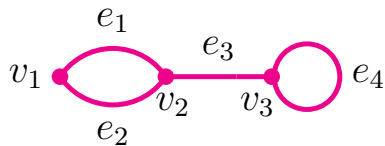
$$A = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

INCIDENCE MATRIX OF A GRAPH

Let G be a graph with vertices $v_1, v_2, \dots, v_n$ and edges $e_1, e_2, \dots, e_m$.
The *incidence matrix* of G is the $n \times m$ matrix $B = [b_{ij}]$ with

$$b_{ij} = \begin{cases} 1 & \text{if edge } e_j \text{ is incident with vertex } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Example.



$$B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Exercise. How can you tell from the incidence matrix whether a graph has parallel edges or loops?